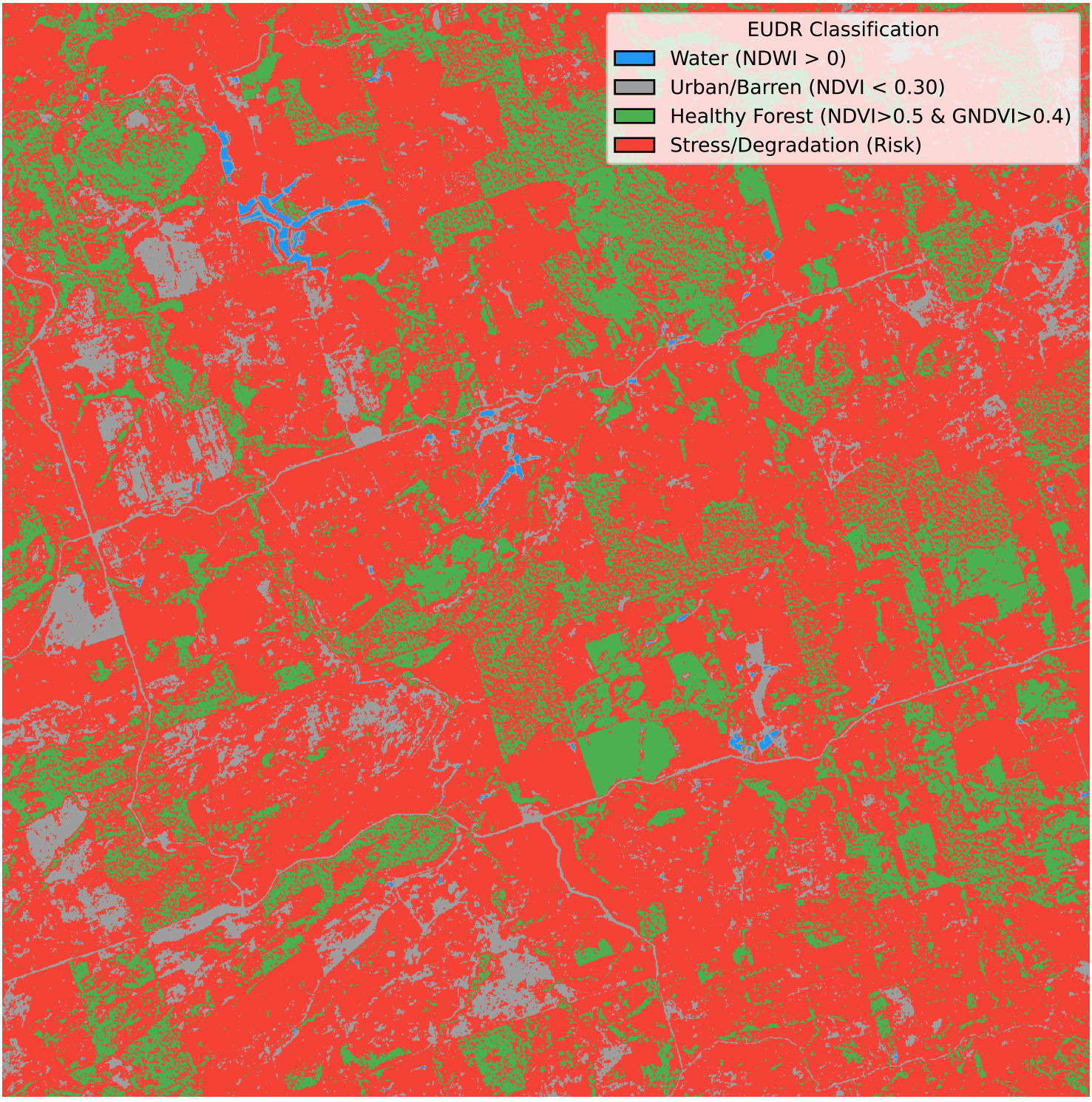Multi-Index Spectral Classification